

Site Ledger V1.0

Role, Access Control & Permission BRD

Document Version: 1.0

Prepared For: Client Review

Project Type: Construction Site ERP System

Platform: Web + Mobile (Android)

1. PURPOSE

1.1 System Objective

Site Ledger ERP का उद्देश्य Construction Business के सभी Financial, Material, Labour, Site Activity एवं Documentation को एक ही Platform पर नियंत्रित करना है।

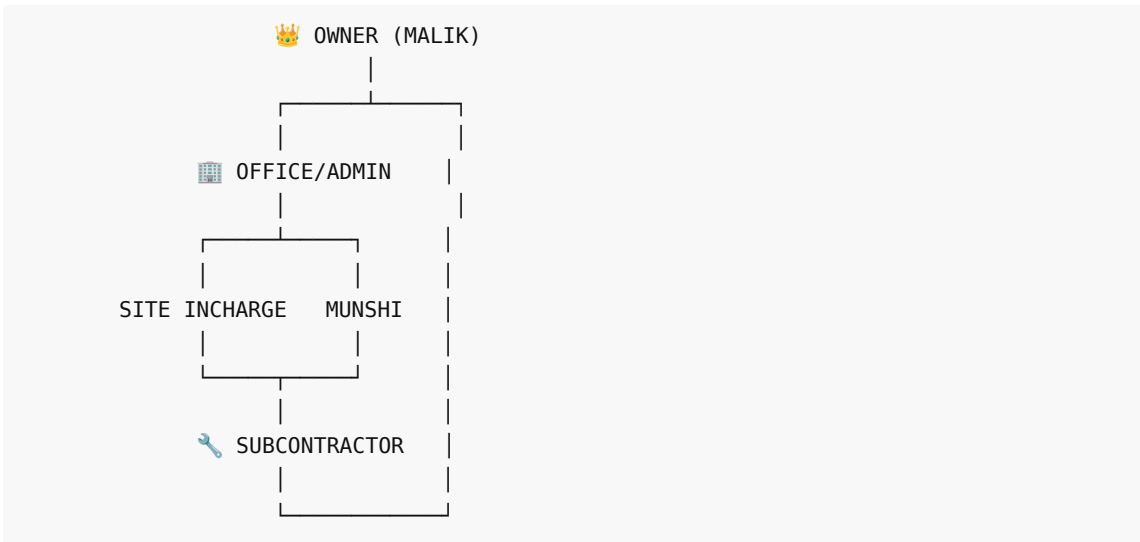
इस BRD का उद्देश्य यह निर्धारित करना है:

- किस User को क्या दिखाई देगा
- किस User को क्या नहीं दिखाई देगा
- कौन क्या Entry कर सकता है
- कौन क्या Edit या Delete कर सकता है

1.2 Core Philosophy

"जिस व्यक्ति को जितना कार्य करना है, उसे केवल उतना ही दिखाई देगा और उतना ही अधिकार मिलेगा।"

2. USER HIERARCHY



2.1 Role Summary

Role	Level	Description
👑 OWNER (Malik)	Level 1	सिस्टम का सर्वोच्च स्तर। पूरे ERP का पूर्ण नियंत्रण।

 OFFICE / ADMIN	Level 2	Owner के बाद सबसे महत्वपूर्ण स्तर। Office ERP की निगरानी करेगा।
 SITE STAFF	Level 3	Site Incharge + Munshi (V1 में दोनों की Base Role समान रहेगी)
 SUBCONTRACTOR	Level 4	Third Party Work Executor

3. ROLE 1: 👑 OWNER (MALIK)

3.1 Description

सिस्टम का सर्वोच्च स्तर। Owner पूरे ERP का पूर्ण नियंत्रण रखेगा।

3.2 Access

Access Type	Details
Sites	✅ सभी Sites
Users	✅ सभी Users
Financial Data	✅ सभी Financial Data
Material Data	✅ सभी Material Data
Reports	✅ सभी Reports
Documents	✅ सभी Documents
Ledgers	✅ सभी Ledgers
Settings	✅ सभी Settings

3.3 Permissions

Permission	Status
User Create	✅
User Edit	✅
User Delete	✅
User Suspend	✅
Site Create	✅
Site Edit	✅
Site Close (Complete/Cancel)	✅
Site Reopen	✅
Any Entry Edit	✅
Any Entry Delete	✅

Permission Assignment	✓
Report Export	✓
System Configuration	✓

3.4 Owner Dashboard

Owner को निम्नलिखित Dashboard दिखाई देगा:

OWNER DASHBOARD			
Total Sites	Running Sites	Completed Sites	Contract Value
12	8	4	₹5,00,00,000
Total Received	Total Expense	Pending Payments	Outstanding Advances
₹3,50,000	₹2,80,000	₹1,50,000	₹25,000
 Overall Profit / Loss : ₹70,000 (Profit)			
Expense Breakdown:			
■ Material Cost ■ Labour Cost ■ Machinery Cost ■ Other Expenses ₹1,50,000 ₹80,000 ₹30,000 ₹20,000			
Site Wise Profit/Loss:			
└─ Site A: +₹25,000		└─ Site D: -₹5,000	
└─ Site B: +₹40,000		└─ Site E: +₹10,000	
└─ Site C: +₹15,000		└─ Site F: -₹15,000	

4. ROLE 2: OFFICE / ADMIN

4.1 Description

Owner के बाद सबसे महत्वपूर्ण स्तर। Office ERP की निगरानी करेगा तथा Field Entries को नियंत्रित करेगा।

4.2 Access Types

Office/Admin के लिए 4 प्रकार के Access हो सकते हैं:

Access Type	Description
 Full Access	सभी Sites पर पूर्ण अधिकार
 Selected Site Access	केवल चयनित Sites पर अधिकार

● View Only Access	केवल देखने का अधिकार (Entry नहीं कर सकता)
● Permission Based Access	Owner द्वारा निर्धारित Module Access

4.3 Configurable Permissions (Owner द्वारा ON/OFF)

Module	Permission	Default
Material Module	View / Add / Edit / Delete	ON
Labour Module	View / Add / Edit / Delete	ON
Expense Module	View / Add / Edit / Delete	ON
Machinery Module	View / Add / Edit / Delete	ON
Documents Module	View / Upload / Delete	ON
Reports Module	View / Export	ON
User Management	View / Create (limited)	OFF
Edit Rights	Edit any entry	ON
Delete Rights	Delete any entry	ON

4.4 Office Responsibilities

Responsibility	Description
Site Monitoring	सभी Sites की नियमित निगरानी
Entry Verification	Field Entries की जांच और सत्यापन
Error Correction	गलत Entries को सही करना
User Management	Users को Manage करना
Report Generation	Reports तैयार करना
Payment Monitoring	Payments की निगरानी

5. ROLE 3: 🧑 SITE STAFF

5.1 Description

इस Role में निम्नलिखित शामिल होंगे:

- **Site Incharge** — साइट का मुख्य प्रभारी
- **Munshi** — साइट पर डाटा एंट्री करने वाला व्यक्ति

V1 में दोनों की Base Role समान रहेगी। इनकी Permission Office/Admin द्वारा नियंत्रित होगी।

5.2 Site Visibility

Rule	Description
Assigned Sites Only	Site Staff केवल उन्हीं Sites को देख सकेगा जो उसे Assign की गई हैं
Other Sites Hidden	उसे अन्य किसी Site की जानकारी नहीं दिखाई जाएगी

5.3 Financial Restrictions

Site Staff को निम्नलिखित जानकारी नहीं दिखाई जाएगी:

Hidden Data	Reason
× Contract Value	Financial confidentiality
× Total Site Expense	Financial confidentiality
× Company Accounts	Financial confidentiality
× Profit/Loss	Financial confidentiality
× Payment Received	Financial confidentiality
× Company Financial Summary	Financial confidentiality

5.4 Permission Matrix (Admin ON/OFF करेगा)

Module	Permissions	Default
 Material Module	☐ View Material ☐ Add Material Entry ☐ Material Shifting ☐ Material Stock View	ON
 Expense Module	☐ View Expense ☐ Add Expense Entry	OFF
 Machinery Module	☐ View Machinery ☐ Add Machinery Entry	OFF
 Labour Module	☐ View Labour ☐ Add Labour Entry	ON
 Document Module	☐ View Documents ☐ Upload Documents	ON
 Balance Module	☐ View Personal Balance ☐ Personal Expense Entry	ON

5.5 Site Staff Entry Types

यदि Permission दी गई हो तो Site Staff निम्नलिखित Entries कर सकेगा:

Material Arrival Entry

Field	Type	Example
Material Name	Text	"OPC 43 Grade Cement"
Quantity	Number	100
Unit	Select	Bags

Vendor	Text	"UltraTech"
Vehicle	Text	"UP14 AB 1234"
Date	Date	01-06-2026
Photo	Image	

Material Shifting Entry

Field	Type	Example
Material Name	Text	"Steel TMT 500"
From Location	Select	"Main Store"
To Location	Select	"Floor 3"
Quantity	Number	50
Vehicle	Text	"Truck No. 5678"
Date	Date	01-06-2026

Machinery Entry

Field	Type	Example
Machine Name	Text	"JCB 3DX"
Hours	Number	8
Rate	Currency	₹1,200/hour
Vendor	Text	"Verma JCB Service"
Date	Date	01-06-2026
Calculation	Auto	$8 \times ₹1,200 = ₹9,600$

Vehicle / Transport Entry

Field	Type	Example
Vehicle Type	Text	"Truck / Tractor / Dumper"
Trips	Number	5
Quantity	Number	25 (Cum)
Rate	Currency	₹2,500/trip
Calculation	Auto	$5 \times ₹2,500 = ₹12,500$

Labour Entry

Field	Type	Example
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Labour Count	Number	10
Mistri Count	Number	3
Total Labourers	Auto	13
Date	Date	01-06-2026
Amount	Currency	₹10,400

💰 Petty Expense Entry

Field	Type	Example
Expense Type	Select	Water / Tea / Tools / Diesel / Misc
Amount	Currency	₹500
Remark	Text	"Drinking water for workers"
Photo	Image	
Payment Source	Select	Company Advance / Personal Money / Vendor Credit

6. 🔒 ENTRY LOCK SYSTEM

6.1 Rule

Site Staff द्वारा Save की गई Entry तुरंत Lock हो जाएगी।

6.2 After Locking

Action	Status
 Edit	× नहीं कर सकता
 Delete	× नहीं कर सकता
 Modify	× नहीं कर सकता

6.3 Correction Process

Site Staff ने गलती की → Site Staff Office को बताएगा
↓
Office Verification करेगा → सही है?
↓
Office Correction करेगा → Entry सही की जाएगी
↓
📄 Audit Log बनेगा → Old Value, New Value, User, Date, Time

7. 📊 MUNSHI / SITE STAFF BALANCE LEDGER

7.1 Purpose

Company Advance, Personal Money तथा Vendor Credit का पूरा हिसाब रखना।

7.2 Balance Dashboard

Site Staff को अपना Personal Balance Dashboard दिखाई देगा:

MUNSHI BALANCE DASHBOARD	
Company Amount Received ₹20,000	Total Expenses Submitted ₹15,000
Current Balance ₹5,000	

7.3 Example Scenarios

Example 1: Company Advance diya

Company ने ₹20,000 दिए
↓
Received = ₹20,000
Expenses = ₹10,000 (किया खर्च)
Balance = ₹10,000 (बचा हुआ - Company को वापस करना है)

Example 2: Personal Money kharch kiya

Munshi ने अपनी जेब से ₹8,000 खर्च किए
↓
Received = ₹20,000
Expenses = ₹28,000 (₹20,000 Company + ₹8,000 Personal)
Balance = -₹8,000
↓
! Meaning: Company को Munshi को ₹8,000 देना बाकी है

7.4 Payment Source

Expense Entry में से एक चुनना होगा:

Source	Description	System Effect
 Company Advance	Company के पैसे से खर्च किया	Company Liability कम होगी
 Personal Money	Munshi ने अपने पैसे से खर्च किया	Company Liability बढ़ेगी (Munshi को पैसे देने होंगे)
 Vendor Credit / Udhar	Vendor से उधार लिया	Vendor Outstanding बढ़ेगा

7.5 Settlement

जब Office/Admin भुगतान करेगा:

- Balance स्वतः Adjust होगा
- Audit Log में दर्ज होगा

8. ROLE 4: SUBCONTRACTOR

8.1 Description

Third Party Work Executor — ठेकेदार जो साइट पर特定 कार्य करता है।

8.2 Access

Access Type	Details
Sites	☒ केवल Assigned Sites
View Data	☒ अपना Work, Quantity, Rate, Payment Status
Hidden Data	× Company Profit, Contract Value, Other Expenses, Company Accounts, Other Contractor Data

8.3 What Subcontractor Can See

SUBCONTRACTOR DASHBOARD	
	Work Done : 500 Sqft Plastering
	Amount Earned : ₹25,000
	Amount Paid : ₹15,000
	Pending Amount: ₹10,000

8.4 Work Entry

Field	Type	Example
Date	Date	01-06-2026
Work Type	Text	"Plastering / PCC / Painting"
Quantity	Number	150
Unit	Select	Sqft / Meter / Cum / Nos
Rate	Currency	₹50/sqft
Payable Amount	Auto	150 × ₹50 = ₹7,500

8.5 Material Supplied Entry

Field	Type	Example
Material Name	Text	"Cement"
Quantity	Number	50
Unit	Select	Bags
Rate	Currency	₹350/bag
Bill Photo	Image	
Total Amount	Auto	$50 \times ₹350 = ₹17,500$

9. AUDIT TRAIL

9.1 Purpose

हर बदलाव रिकॉर्ड होगा:

Audit Field	Description
Old Value	पुरानी Value
New Value	नई Value
User	किसने बदला
Date	कब बदला
Time	कितने बजे बदला

9.2 Data Integrity

कोई भी Record Permanently Delete नहीं होगा।
सभी बदलाव Audit Log में रिकॉर्ड होंगे।

10. SECURITY PRINCIPLES

#	Principle
1	User केवल Assigned Site देखेगा
2	Field Users Financial Data नहीं देखेंगे
3	Entry Save होते ही Lock होगी
4	केवल Office/Admin Edit कर सकेगा
5	सभी बदलाव Audit Log में रिकॉर्ड होंगे
6	Owner को Real-Time Visibility मिलेगी
7	Permission Module आधारित होगी

11. FINAL ROLE SUMMARY

ROLE SUMMARY TABLE		
 OWNER	= Everything – Full System Control	
 OFFICE/ ADMIN	= Configurable Access – Owner sets limits • Full / Selected Site / View Only / Perm	
 SITE STAFF	= Permission Based Entry Access • Assigned Sites Only • Financial Data Hidden • Entries Auto-Locked • Personal Balance Tracking	
 SUBCON- TRACTOR	= Own Work Only • अपना Work, Quantity, Rate, Payment • Company Financials Hidden	

12. MODULE-WISE ACCESS MATRIX

Module	OWNER	OFFICE/ADMIN	SITE STAFF	SUBCONTRACTOR
Owner Dashboard	✓ Full	✓ Full	×	×
Site Dashboard (Financial)	✓ Full	✓ Full	× (Hidden)	×
Site Dashboard (Operational)	✓ Full	✓ Full	✓ Limited	✓ Limited
Sites — All	✓	✓ Configurable	✓ Assigned Only	✓ Assigned Only
Sites — Create/Edit/Delete	✓	✓ Configurable	×	×
Ledger (Accounting)	✓ Full	✓ Configurable	×	×
Materials — View	✓	✓	✓ (If Permitted)	×
Materials — Add Entry	✓	✓	✓ (If Permitted)	×
Materials — Shifting	✓	✓	✓ (If Permitted)	×

Labour — View	✓	✓	✓ (If Permitted)	×
Labour — Add Entry	✓	✓	✓ (If Permitted)	×
Machinery	✓	✓	✓ (If Permitted)	×
Vehicle/Transport	✓	✓	✓ (If Permitted)	×
Petty Expense	✓	✓	✓ (If Permitted)	×
Balance Ledger	✓	✓	✓ Own Only	×
Advances	✓ Full	✓ Configurable	×	×
Payments	✓ Full	✓ Configurable	×	×
Documents — View	✓	✓	✓	✓
Documents — Upload	✓	✓	✓	✓
Subcontractor Work	✓ All	✓ All	×	✓ Own Only
Users — View	✓ All	✓	×	×
Users — Create/Edit	✓ All	×	×	×
Reports	✓ Full	✓ Configurable	×	×
Audit Logs	✓ Full	✓	×	×
System Settings	✓ Full	×	×	×

13. SYSTEM MODULES (V1.0)

13.1 Core Modules

#	Module	Description
M1	Authentication	JWT-based login, role verification
M2	Site Management	Create, edit, manage site lifecycle
M3	Dashboard	Owner + Site dashboards with KPIs
M4	Material Management	Purchase, shift, consume, inventory tracking
M5	Labour Management	Daily labour entry with auto-lock

M6	Machinery Management	Machine hours, rate, vendor tracking
M7	Vehicle/Transport	Vehicle trips, quantity, rate tracking
M8	Petty Expense	Small expense tracking with payment source
M9	Balance Ledger	Munshi/Site Staff personal balance tracking
M10	Ledger / Accounting	Credit/Debit entries, profit/loss calculation
M11	Advance Management	Advance to personnel, expense tracking, settlement
M12	Payment / Billing	Bill tracking, payment received, pending
M13	Document Management	17 document types, version control, photo upload
M14	Subcontractor Work	Work logging, material, payment status
M15	User Management	User CRUD, roles, permissions, suspend
M16	Audit Trail	Complete activity logging
M17	Permission System	Module-level permission ON/OFF per role

13.2 User Count Per Module

Screens / Modules	Owner	Office Admin	Site Incharge	Munshi	Subcontractor	Subcontractor Admin
Login	✓	✓	✓	✓	✓	✓
Owner Dashboard	✓	✓	×	×	×	×
Site List	✓	✓	✓	✓	✓	✓
Site Detail Dashboard	✓	✓	✓ (Limited)	✓ (Limited)	✓ (Limited)	✓ (Limited)
Site Ledger	✓	✓	×	×	×	×
Materials	✓	✓	✓	✓	×	×
Labour	✓	✓	✓	✓	×	×
Machinery	✓	✓	✓ (If Perm)	✓ (If Perm)	×	×
Vehicle/Transport	✓	✓	✓ (If Perm)	✓ (If Perm)	×	×
Petty Expenses	✓	✓	✓ (If Perm)	✓ (If Perm)	×	×
Balance Ledger	✓	✓	✓	✓	×	×
Advances	✓	✓	×	×	×	×

Payments	✓	✓	×	×	×	×
Documents	✓	✓	✓	✓	✓	✓
Subcontractor Work	✓	✓	×	×	✓	✓
Users	✓	×	×	×	×	✓ (Limited)
Audit Logs	✓	✓	×	×	×	×
Settings	✓	×	×	×	×	×

14. TECHNICAL SPECIFICATIONS

14.1 Tech Stack

Layer	Technology
Frontend	React 19 + Vite 8 + Tailwind CSS 4
Backend	Spring Boot 3.2.5 (Java 21)
Database	PostgreSQL 16
Auth	JWT (24hr expiry)
Mobile	Capacitor 8 (Android)
Charts	Recharts 3
HTTP	Axios 1

14.2 Default Login Credentials

Username	Password	Role
owner	owner123	 Owner
admin	admin123	 Office Admin
siteincharge	site123	 Site Incharge
munshi	munshi123	 Munshi
subcontractor	sub123	 Subcontractor
subadmin	subadmin123	 Subcontractor Admin

14.3 Deployment

```
# Backend
cd backend && mvn clean package -DskipTests
java -jar target/site-ledger-backend-1.0.0.jar # Port 8081
```

```
# Frontend
cd frontend && npm install && npm run dev          # Port 5173

# Database
# PostgreSQL on localhost:5444, DB: site_ledger
```

15. 🚀 FUTURE ENHANCEMENTS (V2.0+)

#	Feature	Priority
1	Barcode/QR Scan for material tracking	Medium
2	GPS Check-in for site visits	Medium
3	Offline Support (PWA) for field data entry	High
4	Multi-language (Hindi UI)	Medium
5	Push Notifications for pending tasks	Medium
6	Excel/PDF Report Export	High
7	Email Reports auto-scheduled	High
8	GST Invoicing integration	Low

16. ✅ BUG FIXES APPLIED

#	Issue	Fix
1	locked column missing in labour_entries	Added column to database
2	user_id NULL in Material Transactions	Added setUser() in controller
3	user_id NULL in Labour Entries	Added setUser() in controller
4	user_id NULL in Advances	Added setUser() in controller
5	user_id NULL in Payments	Added setUser() in controller
6	user_id NULL in Documents	Added setUser() in controller